

Network Scanning & Threat Analysis

Nmap Scanning

nmap <target> : runs a scan and returns the Port/Protocol, state and service table.

nmap -O <target> : identifies the operating system on which the target is running.

nmap -sV <target> : returns an additional column , version of the service.

nmap --stats-every <time>s <target> : prints the status for every *time* seconds while running the scan.

```
dhanush@Dhanushs-MacBook-Air ~ % sudo nmap -O -sV --stats-every 20s scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 15:57 +0530
Stats: 0:00:20 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 79.95% done; ETC: 15:58 (0:00:04 remaining)
Stats: 0:00:46 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 83.33% done; ETC: 15:58 (0:00:04 remaining)
Stats: 0:01:00 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.91% done; ETC: 15:58 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Not shown: 990 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    open      smtp
53/tcp    open      tcpwrapped
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
593/tcp   filtered  http-rpc-epmap
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Device type: general purpose|router
Running: Linux 5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 5.0 - 5.14, MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 14 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 67.51 seconds
```

nmap -sV -sC <target> : gives detailed information about each port.

nmap -p <port-num> <target> : runs the scan only for the specified port.

```
dhanush@Dhanushs-MacBook-Air ~ % nmap -p 22,80 -sV -sC scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:06 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.36s latency).

PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
|_ _http-favicon: Nmap Project
|_ _http-title: Go ahead and ScanMe!
|_ _http-server-header: Apache/2.4.7 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 22.49 seconds
```

nmap -T<0-5> <target> : balances the speed and aggressiveness of the scan. T0 is slowest, and almost never misses anything, T5 is the fastest and may miss information or even crash the services.

nmap -F <target> : reduces time of scan, by scanning only the top/widely used 100 ports out of the default 1000 ports.

```
dhanush@Dhanushs-MacBook-Air ~ % nmap -T2 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:15 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.30s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    open       smtp
53/tcp    open       domain
80/tcp    open       http
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
593/tcp    filtered  http-rpc-epmap
9929/tcp   open       nping-echo
31337/tcp  open       Elite

Nmap done: 1 IP address (1 host up) scanned in 443.58 seconds
dhanush@Dhanushs-MacBook-Air ~ % nmap -F scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:32 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Not shown: 93 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    open       smtp
53/tcp    open       domain
80/tcp    open       http
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 11.44 seconds
dhanush@Dhanushs-MacBook-Air ~ % nmap -T5 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:30 +05
Warning: 45.33.32.156 giving up on port because retransmission
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Not shown: 971 closed tcp ports (conn-refused)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    open       smtp
53/tcp    open       domain
80/tcp    open       http
135/tcp    filtered  msrpc
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
593/tcp    filtered  http-rpc-epmap
1034/tcp   filtered  zincite-a
1043/tcp   filtered  boinc
1082/tcp   filtered  amt-esd-prot
1863/tcp   filtered  msnp
1984/tcp   filtered  bigbrother
2196/tcp   filtered  unknown
2394/tcp   filtered  ms-olap2
3370/tcp   filtered  satvid-data1nk
3800/tcp   filtered  pwgpsi
6566/tcp   filtered  sane-port
7004/tcp   filtered  afs3-kaserver
7512/tcp   filtered  unknown
9503/tcp   filtered  unknown
9595/tcp   filtered  pds
9929/tcp   open       nping-echo
16000/tcp  filtered  fmsas
19842/tcp  filtered  unknown
20031/tcp  filtered  unknown
31337/tcp  open       Elite
61900/tcp  filtered  unknown
63331/tcp  filtered  unknown

Nmap done: 1 IP address (1 host up) scanned in 69.13 seconds
```

T2

443.5s & 1000 ports

F

11.44s & 100 ports

T2

69.13s & 1000 ports

nmap <target> -oN file.txt/ -oG file.grep/ -oX file.xml : saves the scan results in a file.

```
dhanush@Dhanushs-MacBook-Air ~ % nmap scanme.nmap.org -oN scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-01 16:45 +0530
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 1.20% done; ETC: 16:47 (0:01:22 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
593/tcp   filtered http-rpc-epmap
9929/tcp  open  nping-echo
31337/tcp open  Elite
```

```
UW PICO 5.09                                     File: scan.txt

# Nmap 7.98 scan initiated Sun Mar  1 16:45:49 2026 as: nmap -oN scan.txt scanme.nmap.org
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Not shown: 990 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
593/tcp   filtered http-rpc-epmap
9929/tcp  open  nping-echo
31337/tcp open  Elite

# Nmap done at Sun Mar  1 16:47:00 2026 -- 1 IP address (1 host up) scanned in 71.13 seconds
```

Observations

- **Open ports** : 22, 25, 53, 80, 135, 139, 445, 593, 9929, 31337.
- **Services** : ssh, smtp, domain, http, msrpc, netbios-ssn, microsoft-ds, http-rpc-epmap, nping-echo, Elite
- Nmap runs the “**tcp**” protocol by default.
- **T2** is much **slower** than **T5**. But T5 trades off accuracy for speed.
- **F** managed to maintain accuracy and speed, as it focuses on only the **top 100** ports.
- **-sV** argument gave the **version** of service, while **-sC** gave **details** about each service.
- **-p** argument can be used to focus only on **specific** ports.
- The target system is running on a **Linux 5.X** which was found using **-O** argument.
- The scan output can be written into files using **oN/oG/oX** arguments. These files(*.txt/.grep/.xml*) can help in analysis of the scan in the future.

VirusTotal Intelligence Gathering

What is VirusTotal used for?

VirusTotal helps in identifying how vulnerable the target is. It is used to track any kind of threats(stats) to the target and display them. It also maintains the history of the target which enables analysis of resolved threats as well.

How does passive intelligence differ from active scanning?

Passive intelligence is more like looking at an existing case file. Active scanning is like getting into the field, making the case file. Passive intelligence is made up of the reports given by users, and the developer community. Active scanning includes finding the vulnerabilities in the moment.

How do the VirusTotal findings support or complement nmap results?

```
--- URL ANALYSIS ---  
Detection Stats: {'malicious': 0, 'suspicious': 0, 'undetected': 29, 'harmless': 65, 'timeout': 0}  
Verdict: likely safe.. (0.0%)
```

The VirusTotal findings show that the target is safe and detects no malicious and suspicious stats. So, the open ports in the nmap scan, maybe protected in a way that no one can exploit them, even when they are a vulnerability.

Port/Service Analysis & Vulnerability Reasoning

Port	SERVICE	VERSION
22	ssh	OpenSSH 6.6.1p1
25	smtp	
53	domain	
80	http	Apache httpd 2.4.7(ubuntu)
9929	nping-echo	Nping echo

SSH - ssh enables remote access into the system. Leaving this port open might enable attackers to access the system.

SMTP - smtp is the protocol that delivers mail across the systems. Leaving this port open might enable attackers to pose like authorized personnel, and communicate through the target system.

Domain - DNS translated the names to ip addresses. Leaving this port open enables the attackers to perform DNS enumeration and DNS amplification attacks.

HTTP - Http protocol transfers web pages and resources on the web app. Leaving this port open enables the attackers to perform sql injection, exploit web page vulnerabilities.

Nping echo - Used to perform scans on the target network. Leaving this port open might enable attackers to scan and study the target and plan further attacks. *Without the nping echo port open, the analysis done so far would not be possible.*

VirusTotal

Reputation Details

```
import requests, base64
from google.colab import userdata
API_KEY = userdata.get("API_KEY")
base_url = "https://www.virustotal.com/api/v3"

def getData(url):
    vt_url = f"{base_url}/{url}"
    headers = {
        'x-apikey' : API_KEY
    }
    response = requests.get(vt_url, headers=headers)
    if(response.status_code==200):
        return response.json()
    print("Error : Couldn't fetch data")
    return None
```

```

▶ url = "scanme.nmap.org"
url = base64.urlsafe_b64encode(url.encode()).decode().strip("=")
data = getData(f"urls/{url}")

attributes = data["data"]["attributes"]

stats = attributes["last_analysis_stats"]

label, score = calculate_confidence(stats)

print("\n--- Reputation ---")
totalAgents = sum(stats.values())
print("Reputation score: ", attributes.get("reputation"), " / ", totalAgents)
for stat in stats:
    print(stat, " : ", stats[stat])
print("Verdict:", label)

...
--- Reputation ---
Reputation score:  0 / 95
malicious   : 0
suspicious  : 0
undetected  : 32
harmless    : 63
timeout     : 0
Verdict: likely safe..

```

Domain and IP intelligence

```

print("Domain and IP Intelligence")
print("Domain : ", attributes.get("url"))
print("Title : ", attributes.get("title"))
print("First seen: ", attributes.get("first_submission_date"))
print("Last seen: ", attributes.get("last_submission_date"))

Domain and IP Intelligence
Domain : http://scanme.nmap.org/
Title : Go ahead and ScanMe!
First seen: 1334825457
Last seen: 1773189166

```

Historical/Context Data

```
import datetime

print("Historical Data")

first_submission_timestamp = attributes.get("first_submission_date")
first_submission_date = datetime.datetime.fromtimestamp(first_submission_timestamp)
print("Creation Date : ", first_submission_date.strftime('%Y-%m-%d'))

last_modification_timestamp = attributes.get("last_modification_date")
last_modification_date = datetime.datetime.fromtimestamp(last_modification_timestamp)
print("Last Modified Date : ", last_modification_date.strftime('%Y-%m-%d'))

print("Contextual Data: \nLast scan agents:")
for agent in attributes["last_analysis_results"]:
    print(agent, end = ", ")

Historical Data
Creation Date : 2012-04-19
Last Modified Date : 2026-03-11
Contextual Data:
Last scan agents:
Acronis, Abusix, ADMINUSLabs, Lionix, Criminal IP, AILabs (MONITORAPP), AlienVault, alphaMountain.ai, AlphaSOC, Antiy-AVL, ArcSight
```

Any security risks observed?

```
malicious : 0
suspicious : 0
undetected : 32
harmless : 63
timeout : 0
```

The target has 63 harmless records in the last analysis. The 32 undetected records might be a risk, but they are pretty much ambiguous as they were undetected.

Conclusion

Through this assignment, I explored various ways to scan a target network using the nmap tool. I understood how scanning methods differ, and when to use which. This assignment made me explore basic networking concepts, understand protocols, and how they can turn into a vulnerability.

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